

Assessing uncertainties in satellite-based estimation of evapotranspiration

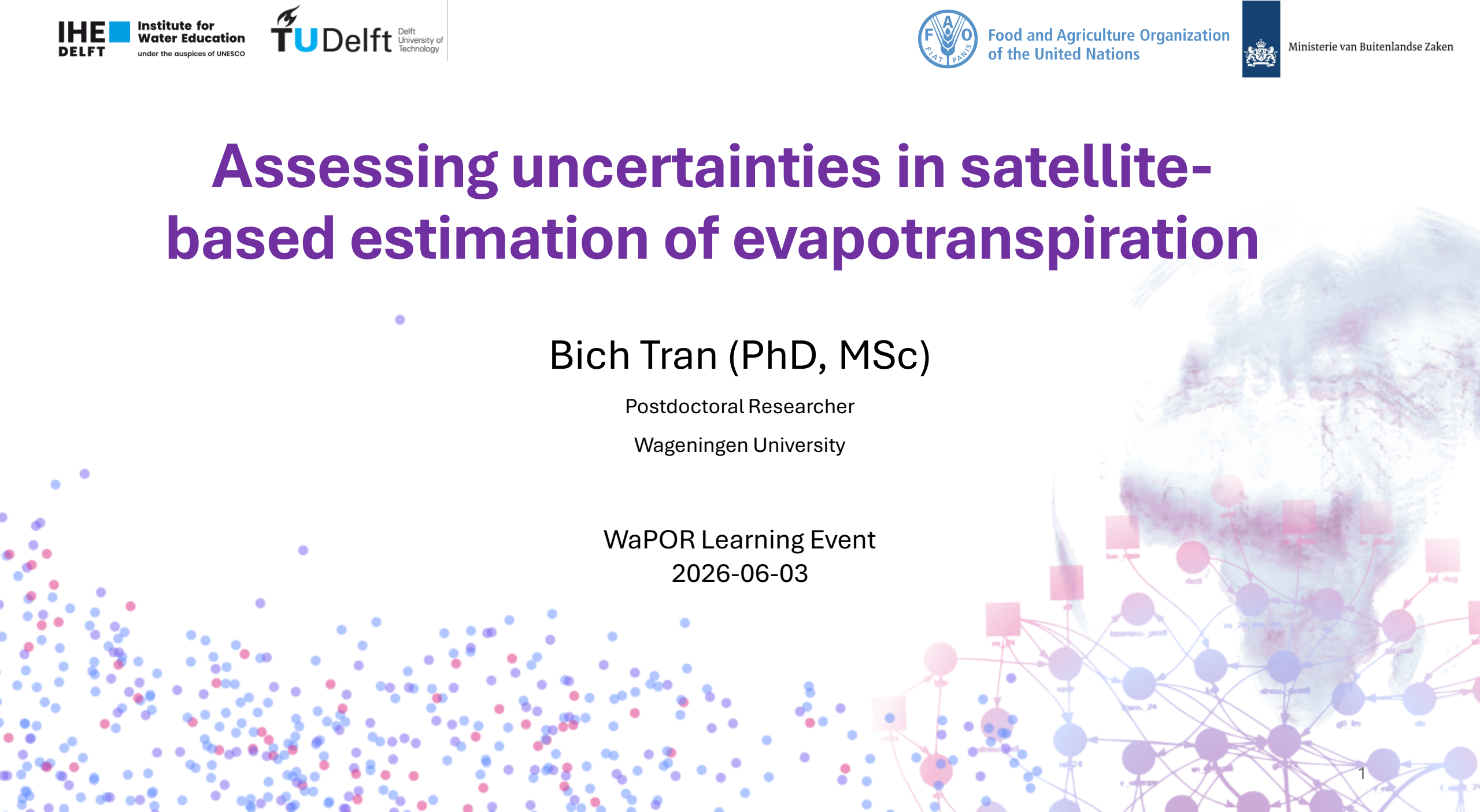
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Postdoctoral Researcher

Wageningen University

WaPOR Learning Event

2026-06-03



About me

- My name is pronounced as /Bik/
- I am currently a researcher at Wageningen University
- Academic backgrounds: Environmental sciences, water resources, hydrology
- Research experiences: Wetlands Ecohydrology, Applied remote sensing, Hydrological modelling, Water accounting, Uncertainty assessment
- Research interests: Changes in the water cycle(s) and the connections to society.



2012-2015 BSc
in Water –
Environment –
Oceanography,
Vietnam

2016-2018 MSc
in Water
Resources
Engineering,
VUB & KUL,
Belgium

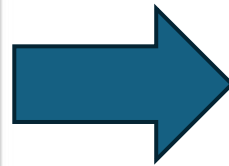
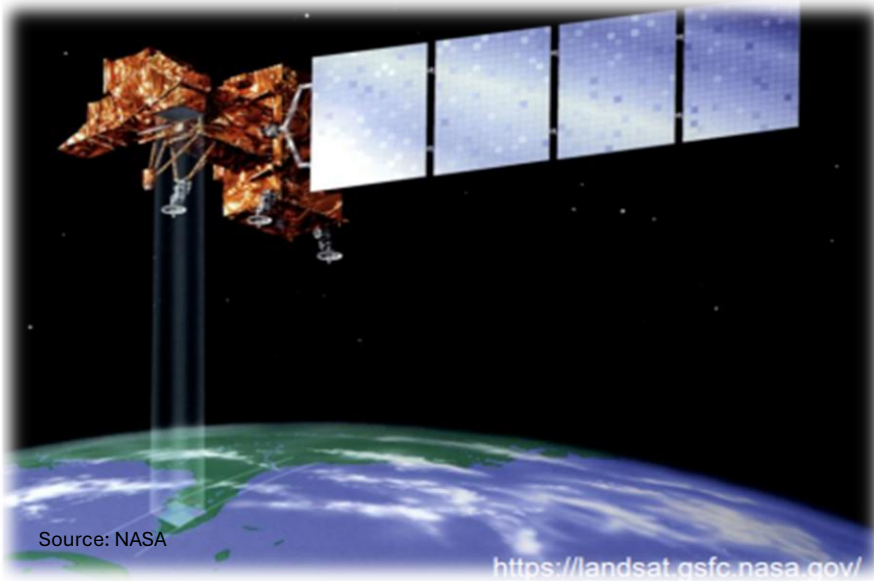
2019-2021
Research
Assistant, IHE
Delft, NL

2021-2026 PhD
in Hydrology &
Water
resources, IHE
Delft & TU
Delft, NL

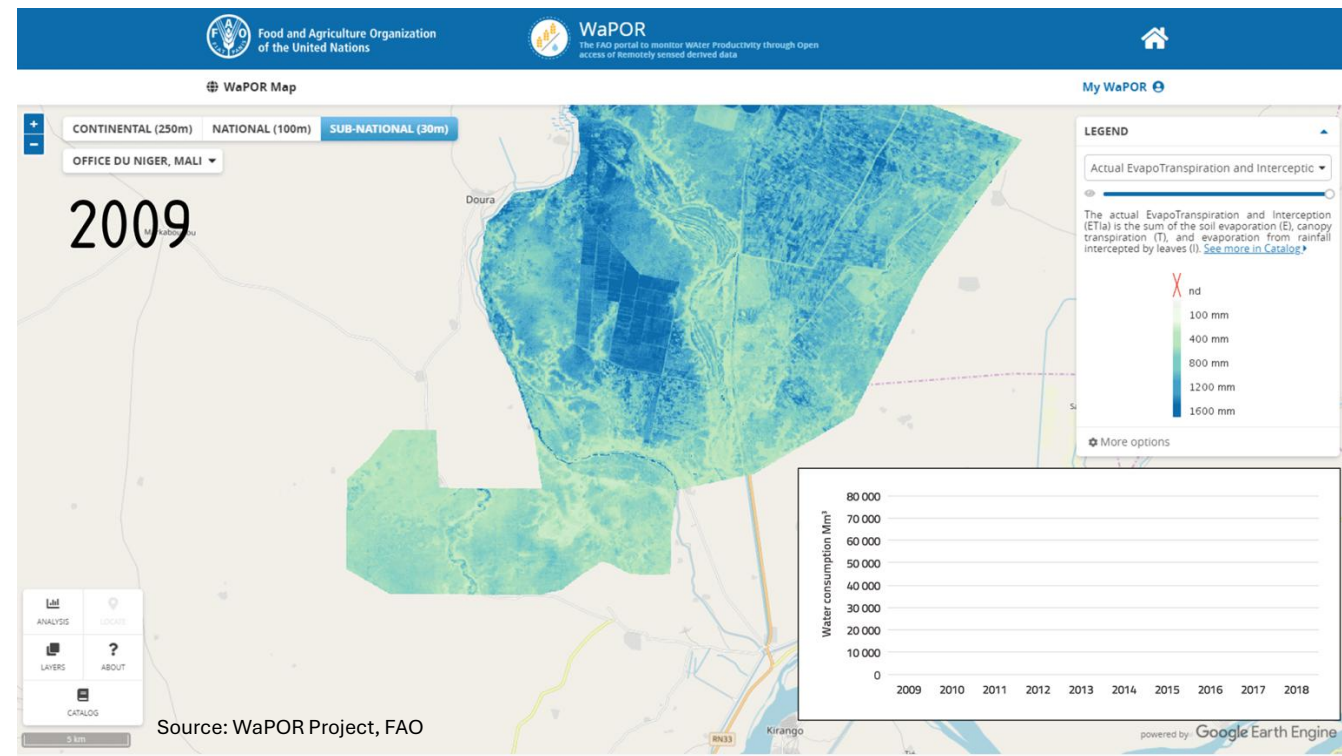
Research Background

Satellite remote sensing is increasingly used to acquire data and information of ET over large areas (e.g., continents, river basins, irrigation schemes)

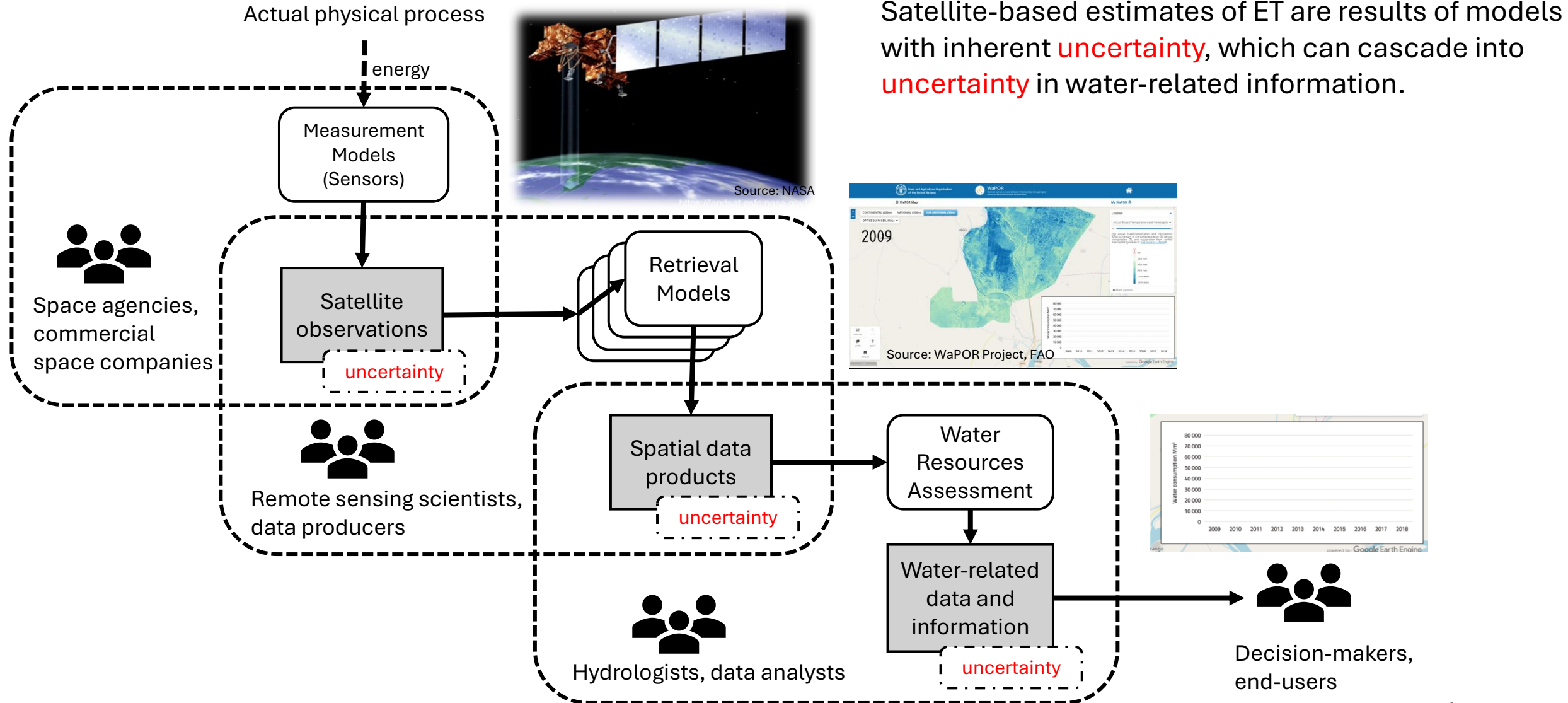
Satellite observations



Spatial ET data products

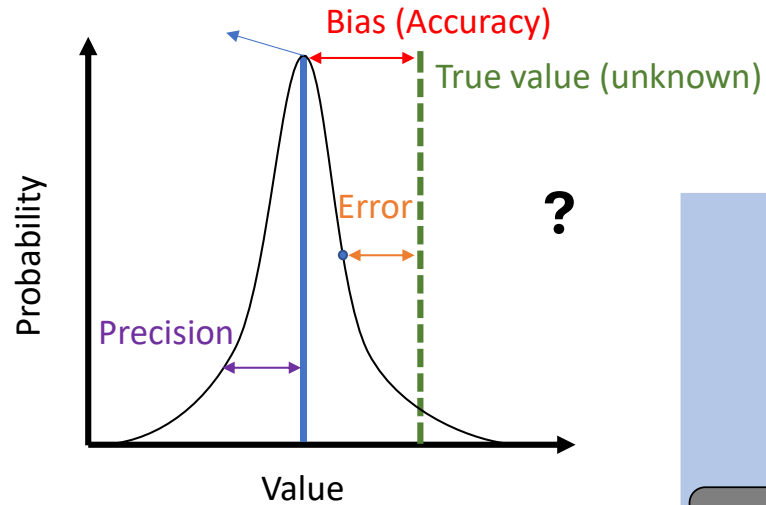


Research problem

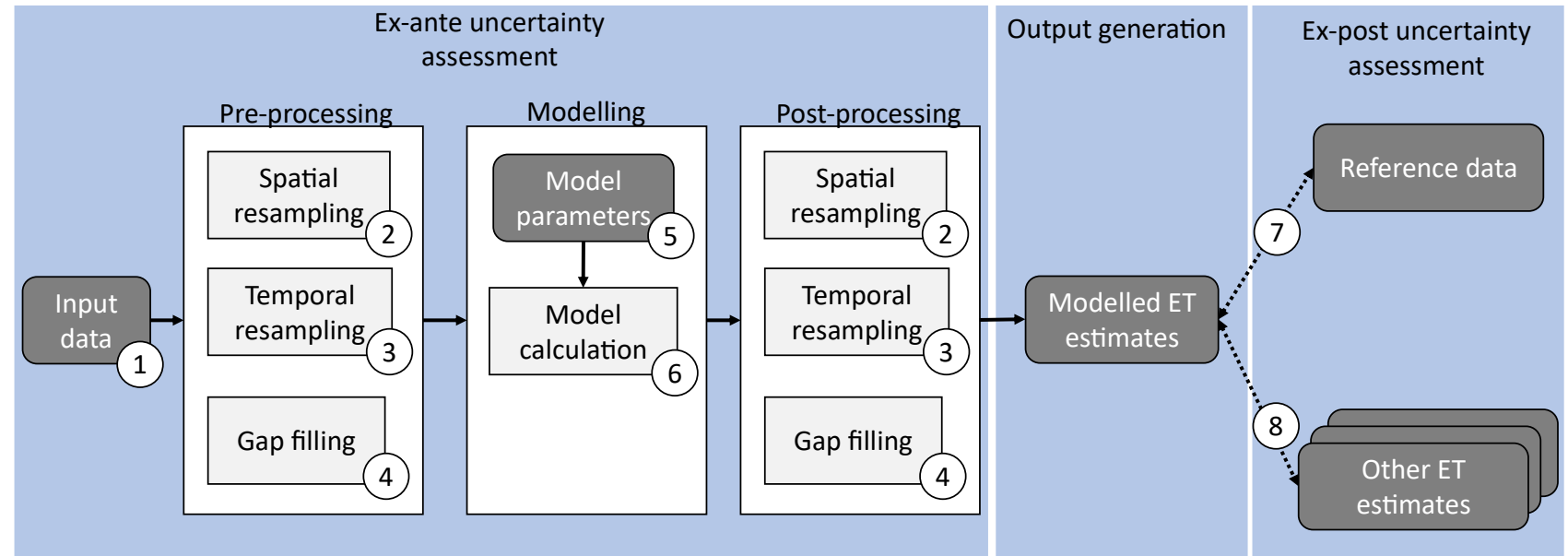


Concepts of uncertainty

Mean of measured values



The terms ‘accuracy’, ‘bias’, and ‘precision’ are used to characterize uncertainty as a probability distribution of errors (**technical uncertainties**)



Sources of uncertainty

1: Input data

2: Change in spatial scale

3: Change in temporal scale

4: Gap filling

5: Model parameterization

6: Model conceptualization

7: Compound uncertainty

8: Relative uncertainty

Raw satellite imagery undergoes a chain of pre- and post-processing and modelling to generate ET data, but introducing ambiguity in which techniques/data sources (**methodological uncertainties**)

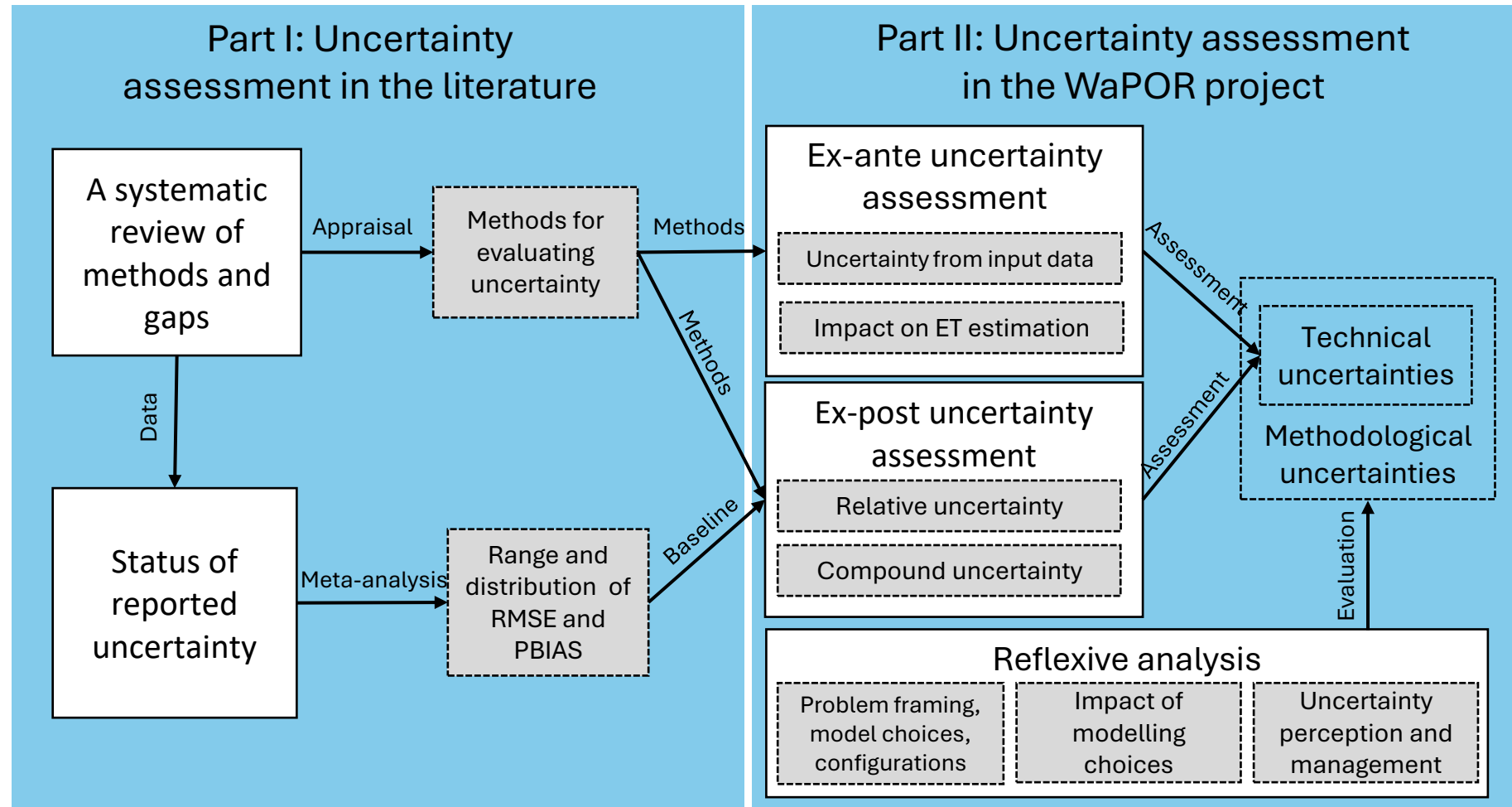
My research objectives

This research aims to enhance the **understanding of both technical and methodological uncertainties** in satellite-based estimation of actual ET and further explore **the implications of these uncertainties**.

Objective 1 Identify how uncertainties of satellite-based ET estimates have been evaluated and the status of uncertainty reported.

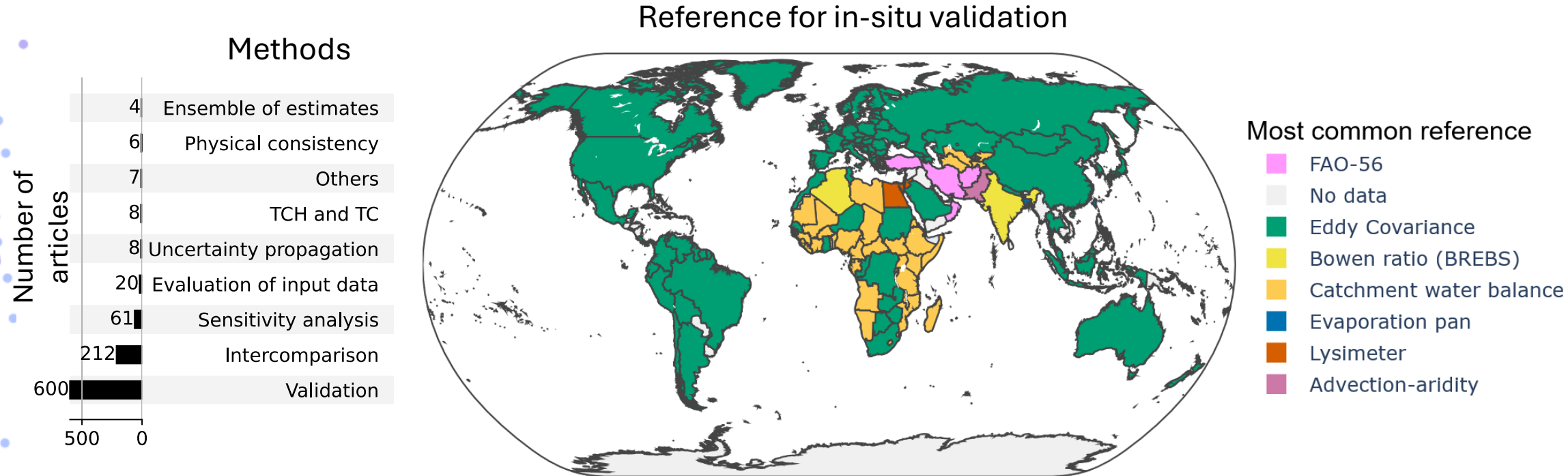
Objective 2 Evaluate the technical uncertainties of a satellite-based ET data product both before (ex-ante) and after (ex-post) data product generation.

Objective 3 Explore the implications of uncertainties in a satellite-based ET data product for its applications.



Uncertainty assessment in the literature

Review of uncertainty assessment methods from 676 research articles



Tran et al. (2023). <https://doi.org/10.5194/hess-27-4505-2023>

A systematic
review of
methods and
gaps

Conclusion

Methods for technical uncertainty assessments are diverse and constrained by the availability of the reference.

Uncertainty assessment in the literature

Meta-analysis of 348 reviewed articles that validate satellite-based ET against Eddy Covariance (EC) measurements

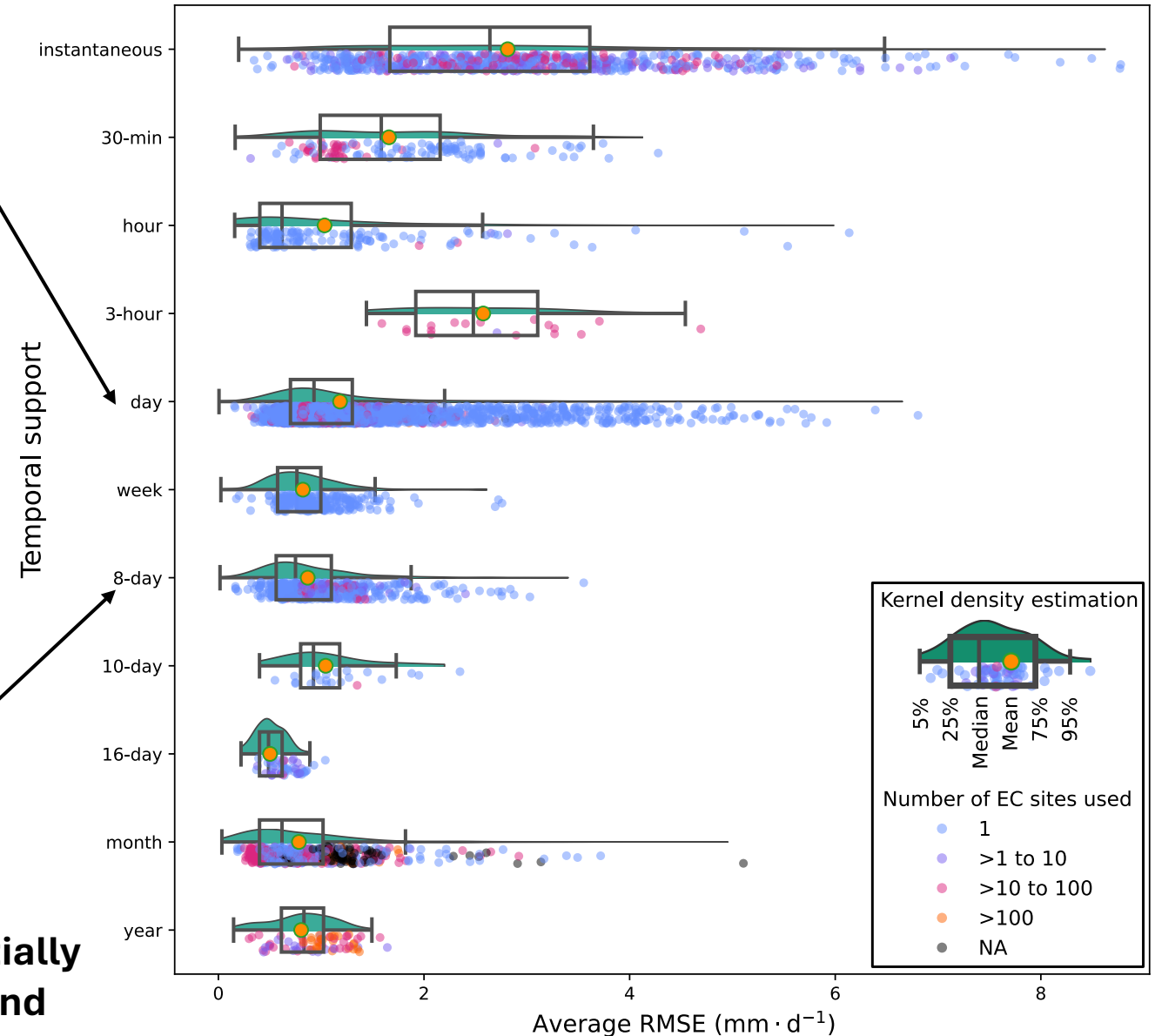
Wide range of uncertainty (RMSE) at daily scale

Smaller range of uncertainty (RMSE) at 8-day scale

Status of reported uncertainty

Conclusion

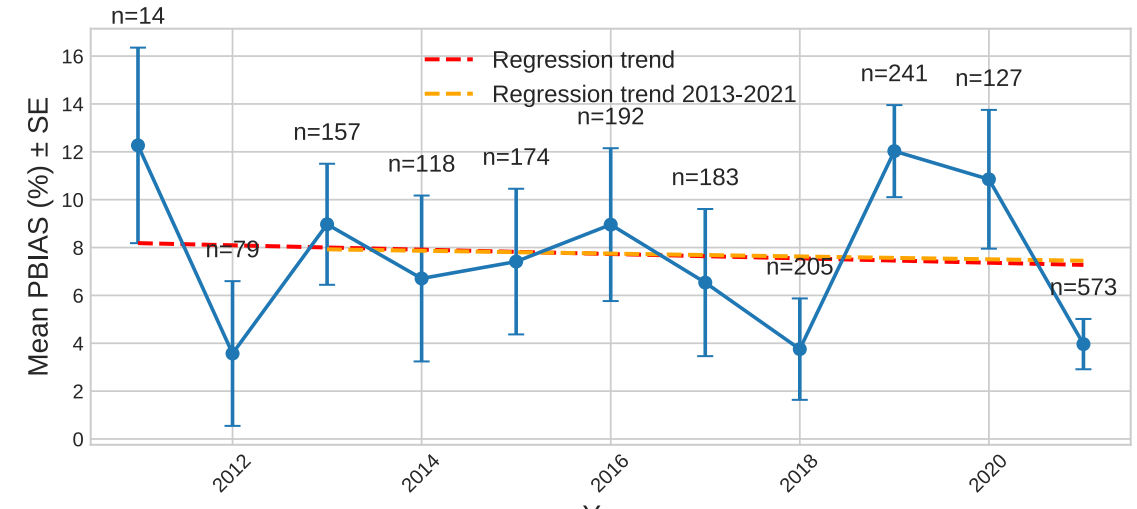
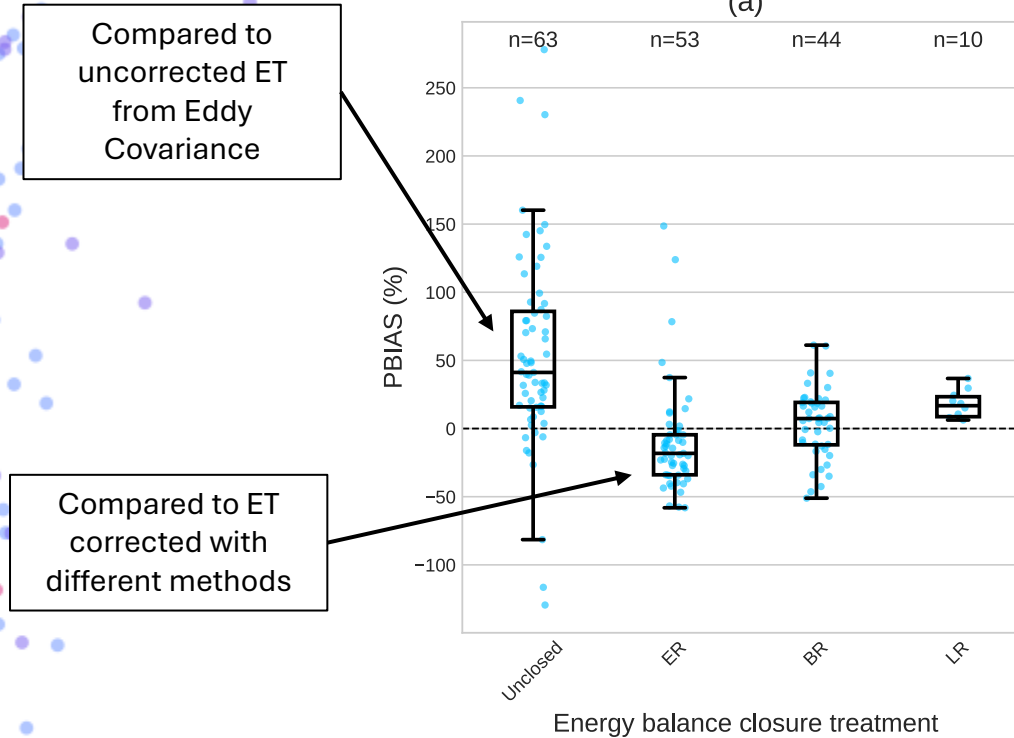
Reported uncertainty varies substantially with scales (e.g., temporal support) and reference (e.g., number of sites)



Tran et al. (2023).

<https://doi.org/10.5194/hess-27-4505-2023>

Uncertainty assessment in the literature



Mean and standard deviation of PBIAS in satellite-based ETa estimates by the year of publication

PBIAS (%) in satellite-based ETa estimates categorized by treatment of energy balance closure

Tran (2026), Chapter 3.

<https://doi.org/10.4233/uuid:6746c17f-dbe3-431c-bb3d-5aabfb3614e0>

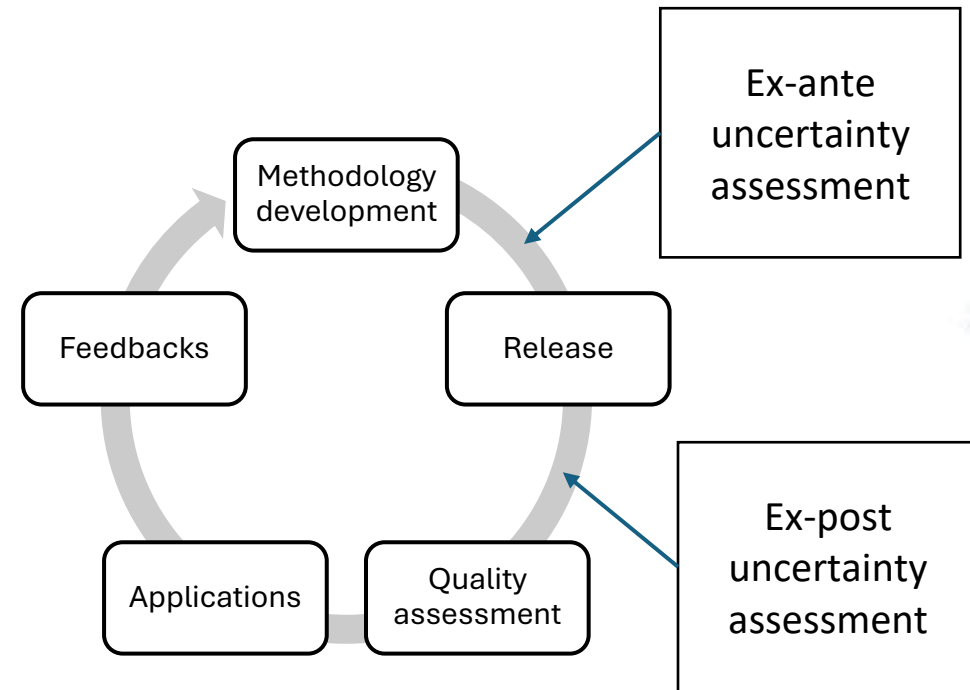
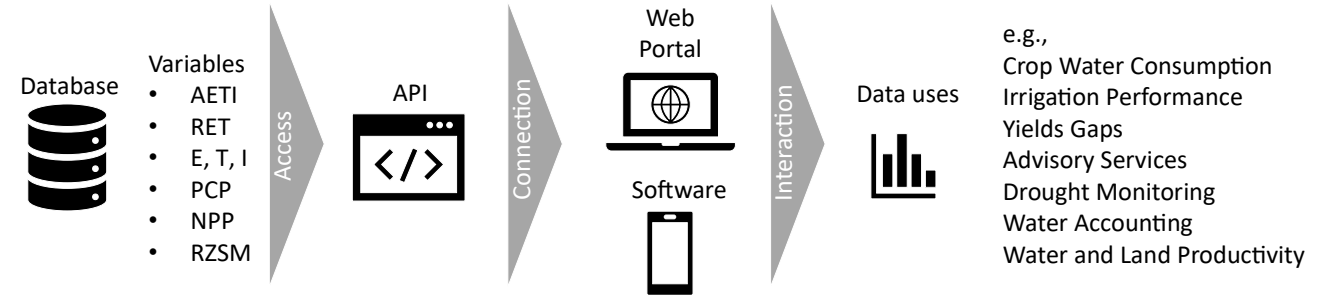
Status of reported uncertainty

Conclusion

The result of in-situ validation depends on the quality of in-situ measurement (up to 50-100% difference in PBIAS). Overall, reported PBIAS has not been significantly improved over a decade of publications.

Uncertainty assessment in the WaPOR project

FAO developed a **publicly accessible near real-time** database using satellite data that allows the monitoring of agricultural **water productivity** at **different scales**.



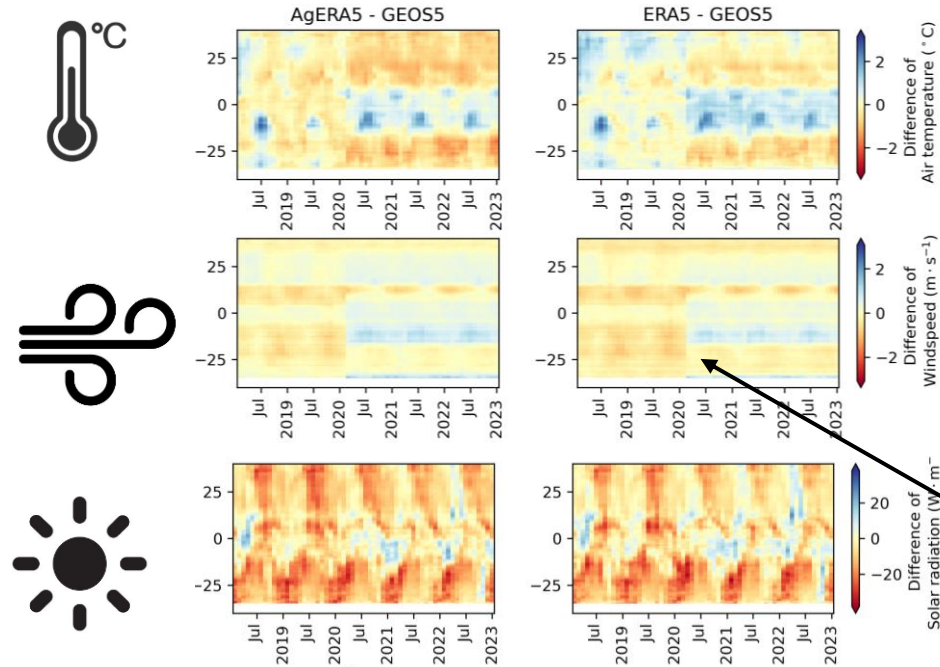
The global level (300m resolution) that covers the entire globe.

The continental and national / river basin level (100 m ground resolution) Northern and sub-Saharan Africa and the Near East (roughly a square of -30W, -40S, 65E, 40N)

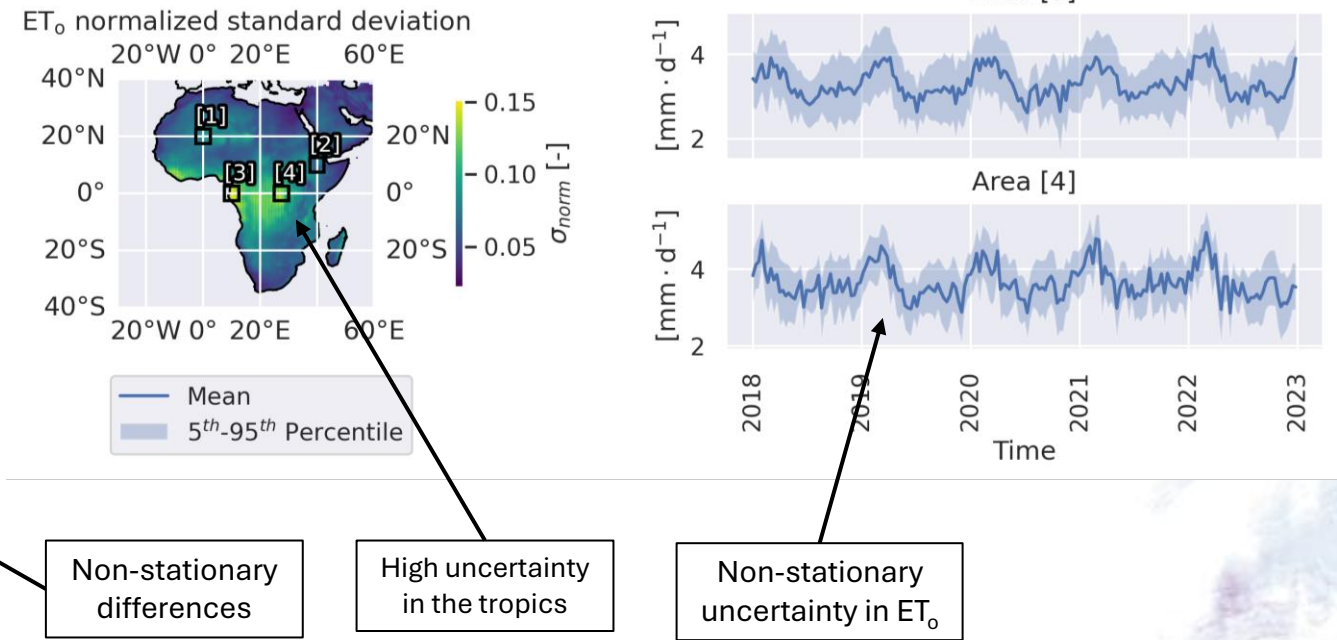
The irrigation scheme and sub-basin (20 m ground resolution) available for these areas:

Technical uncertainties

Intercomparison of meteorological inputs from reanalysis products



Uncertainty in ET of a reference crop (ET_0), resulting from propagating uncertainty in meteorological inputs



Tran et al. (2026) <https://doi.org/10.1080/02626667.2025.2600682>

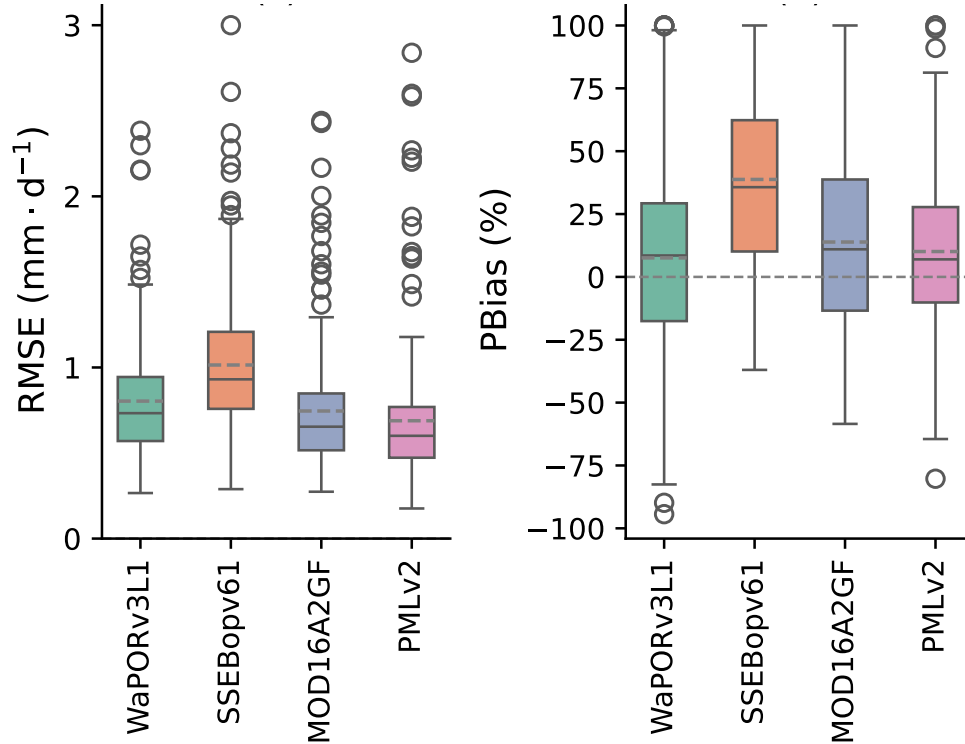
Ex-ante
uncertainty
assessment

Conclusion

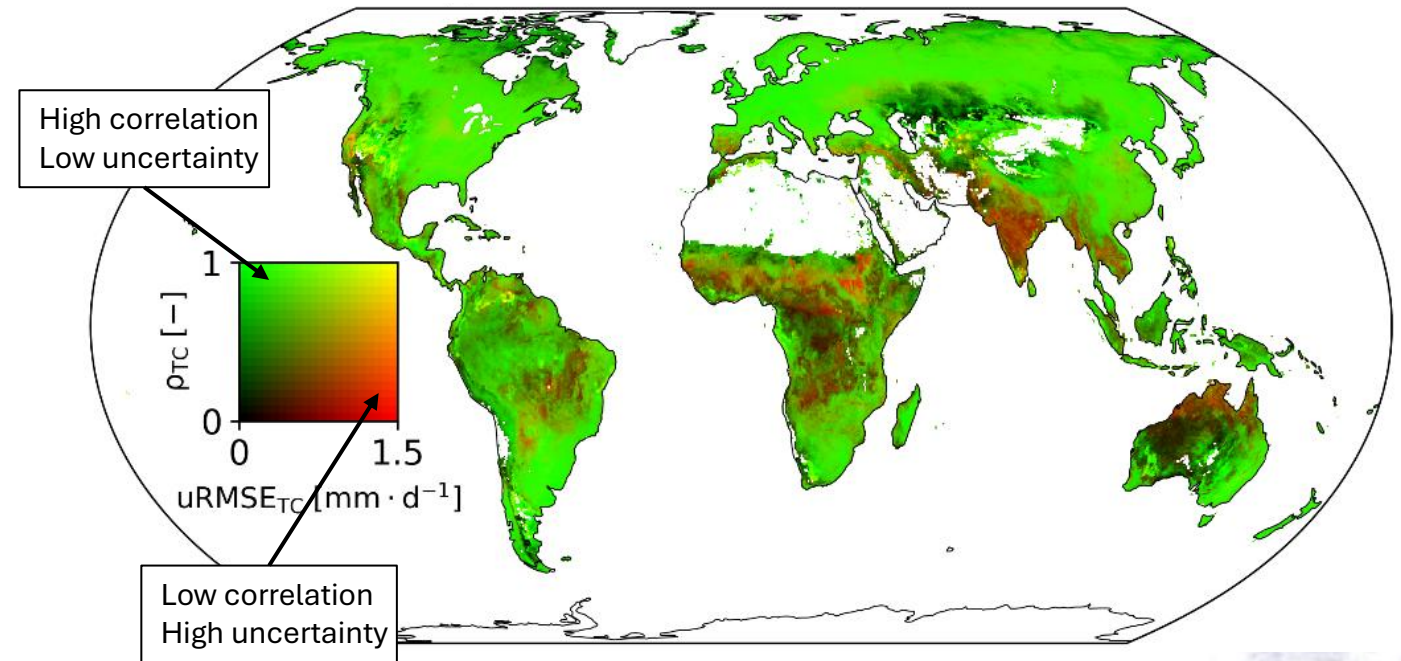
The uncertainty in meteorological inputs varies spatially and temporally, which influences the distribution of uncertainty in ET estimates.

Technical uncertainties

Compound uncertainty evaluated by in-situ validation at 222 EC sites



Relative uncertainty evaluated by triple collocation with three satellite-based ET datasets

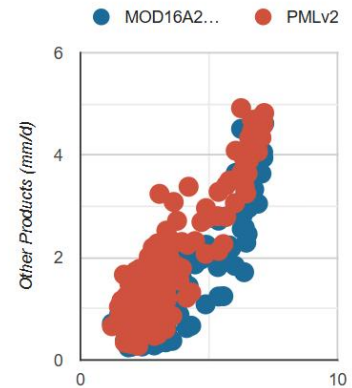
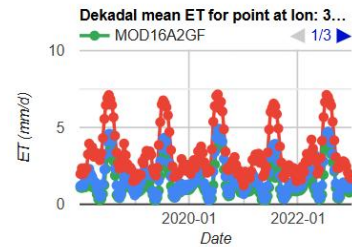
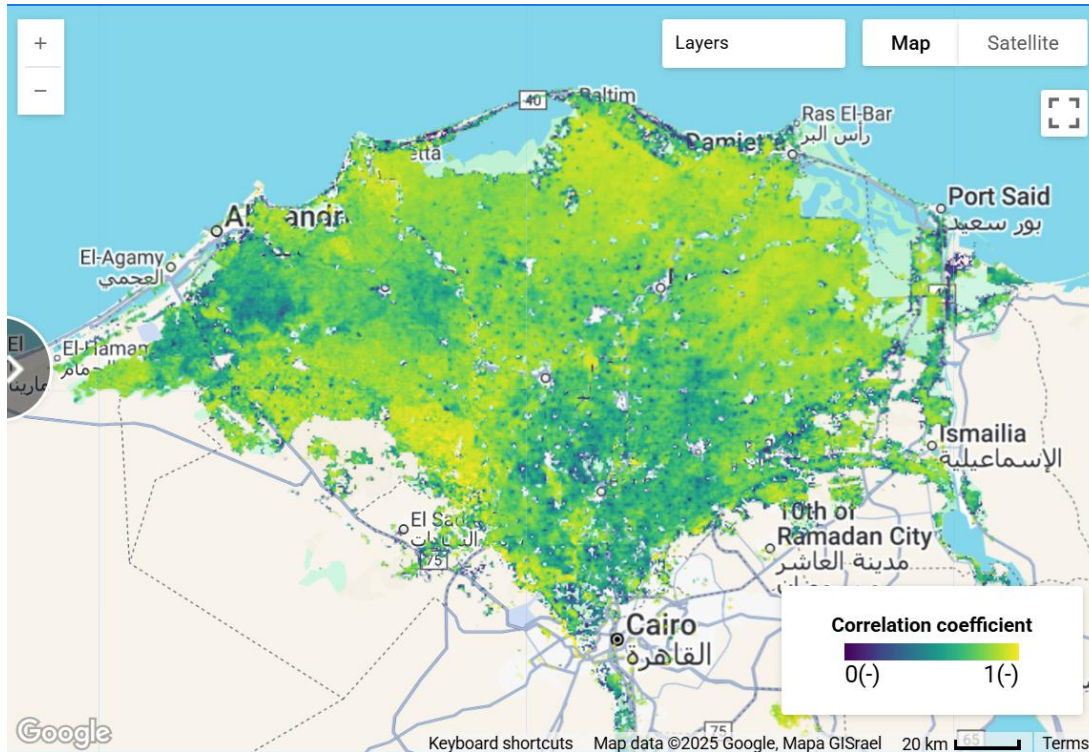


Ex-post
uncertainty
assessment

Conclusion

The uncertainty in ET estimates is comparable for several products at validation sites. Yet, there is still large uncertainty in dry and tropical climates.

Technical uncertainties



Temporal average of correlation coefficient of dekadal ETa from WaPORv3L1 calculated by triple collocation analysis with MOD16A2GF and PMLv2.

Tran (2026), Chapter 5.

<https://doi.org/10.4233/uuid:6746c17f-dbe3-431c-bb3d-5aabfb3614e0>

The 300m-resolution maps can be explored at:

<https://ee-wapor.projects.earthengine.app/view/tca>

Ex-post
uncertainty
assessment

Conclusion

Spatially distributed estimation of relative uncertainty could provide complementary information when the ET estimates are used in comparative analyses such as land and water productivity gap analyses.

Methodological uncertainties

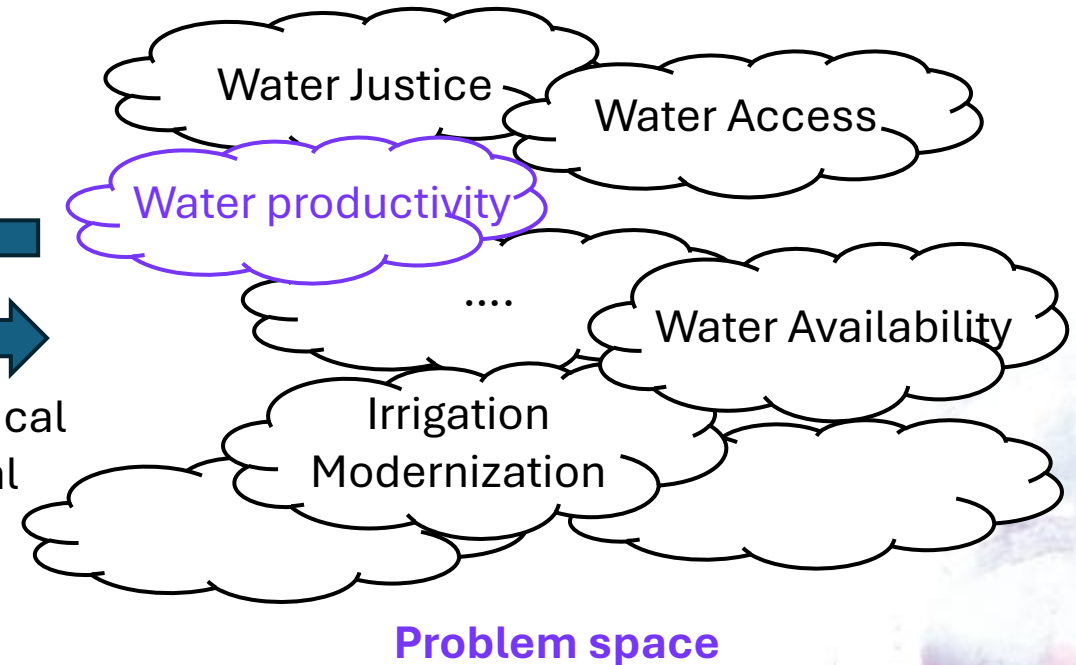
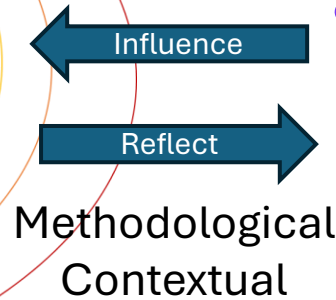
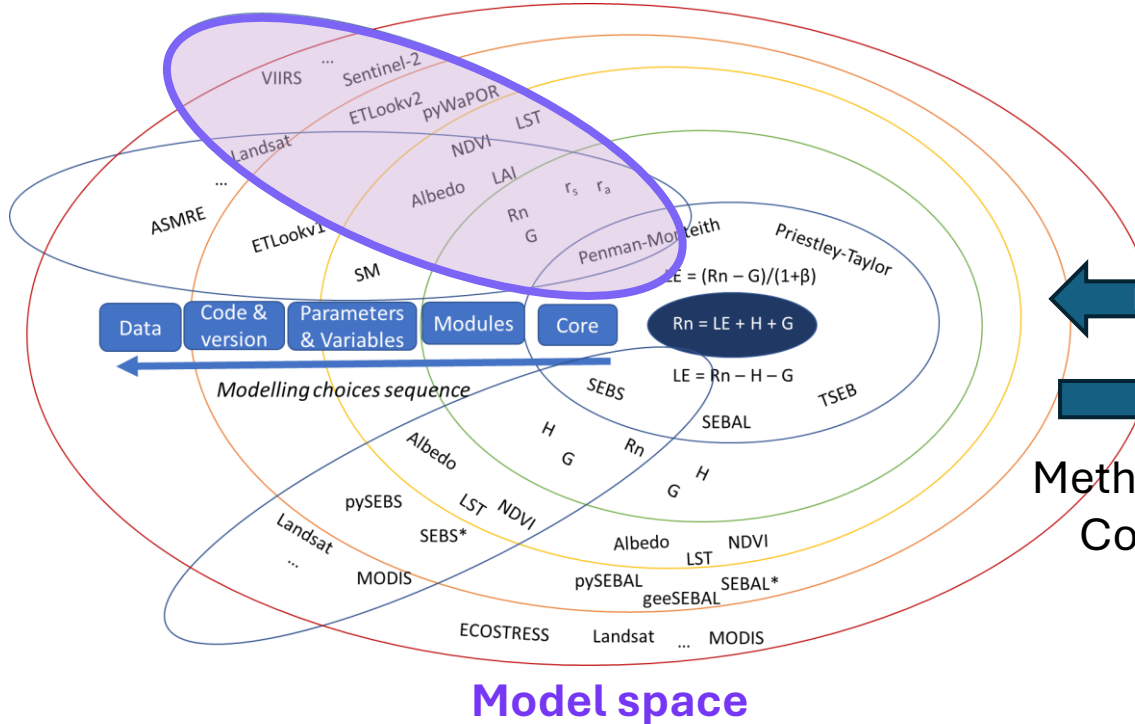
Modelling choices:

- which method?
- which model?
- which data?

But also...

- how the problem are defined,
- which questions are (not) asked, by whom
- who are (not) involved.

WaPOR-ETLook



Reflexive
analysis

Conclusion

Uncertainty is also about how we frame problems of estimation (ambiguity), who we involve and how transparently we communicate, which has implications for water management.

Tran (2026), Chapter 6.

Recommendations

- The uncertainty assessment framework helps evaluate the quality of satellite-based data product before and after data are generated and used in specific applications.
- Refer to quality assessment reports of WaPOR data products for general assessment. But pay attention to the context-specific results (e.g., per land cover, per climate type).
- For specific applications, determine which uncertainty metrics are important for the conclusion and the error budget (i.e., range of uncertainty metrics that are acceptable). This depends greatly on the problem and context that users need data for.

Recommendations

- Technical uncertainty assessments partially provide estimates of uncertainty, contingent on methods and context.
 - Report methods used for uncertainty assessment
 - Evaluate the quality of reference data (e.g., in-situ measurements).
 - Use multiple datasets and methods to estimate relative uncertainty

Recommendations

- Uncertainty assessment should not be a one-off technical step, but an ongoing practice in collaboration with end-users.
- Understand how end-users make decisions and what information and data they use.
- Assimilate satellite-based data and its estimated uncertainty in existing decision framework, in combination with other sources of data.

Take-Home Messages Questions

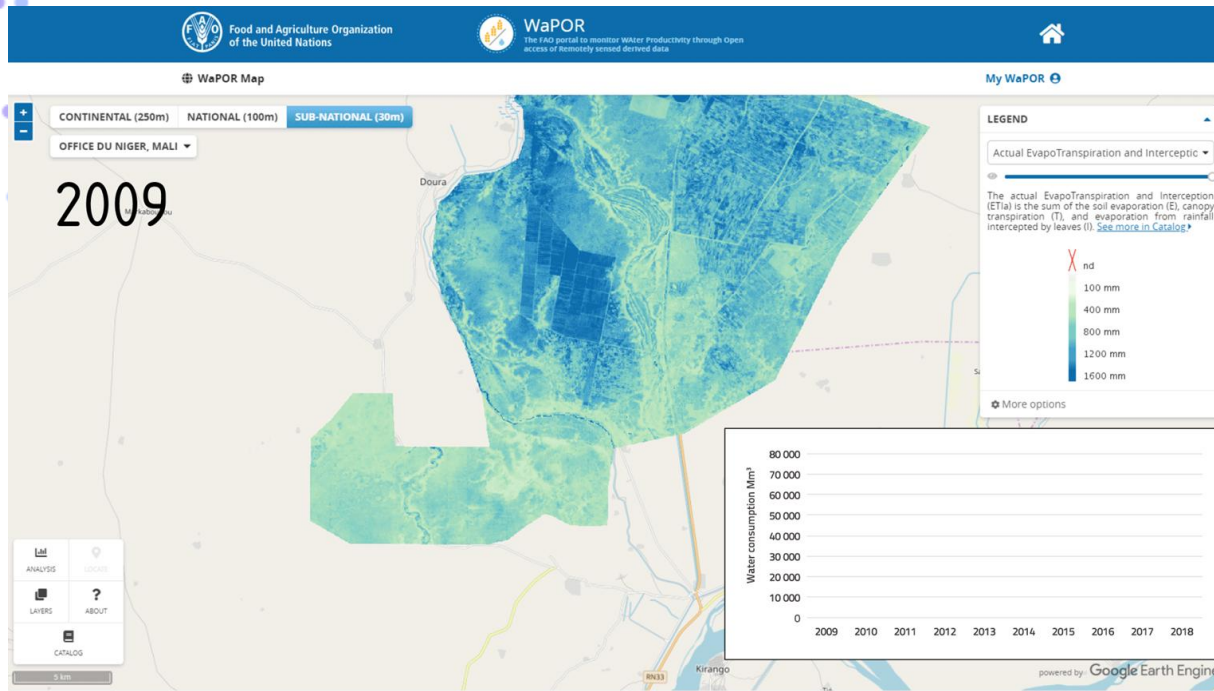
If we cannot eliminate uncertainties

What do you want to use spatial ET data products (e.g., WaPOR) for?

Which quantity needs to be known?

What decisions/conclusions to be made based on that?

How **accurate** and **precise** do the ET data need to be to make a good decision?
(e.g., in terms of percentage bias, mean absolute percentage error)



Mean of measured values

